

[MS-SAMLPR]:

Security Assertion Markup Language (SAML) Proxy Request Signing Protocol

Intellectual Property Rights Notice for Open Specifications Documentation

- **Technical Documentation.** Microsoft publishes Open Specifications documentation ("this documentation") for protocols, file formats, data portability, computer languages, and standards support. Additionally, overview documents cover inter-protocol relationships and interactions.
- **Copyrights.** This documentation is covered by Microsoft copyrights. Regardless of any other terms that are contained in the terms of use for the Microsoft website that hosts this documentation, you can make copies of it in order to develop implementations of the technologies that are described in this documentation and can distribute portions of it in your implementations that use these technologies or in your documentation as necessary to properly document the implementation. You can also distribute in your implementation, with or without modification, any schemas, IDLs, or code samples that are included in the documentation. This permission also applies to any documents that are referenced in the Open Specifications documentation.
- **No Trade Secrets.** Microsoft does not claim any trade secret rights in this documentation.
- **Patents.** Microsoft has patents that might cover your implementations of the technologies described in the Open Specifications documentation. Neither this notice nor Microsoft's delivery of this documentation grants any licenses under those patents or any other Microsoft patents. However, a given Open Specifications document might be covered by the Microsoft [Open Specifications Promise](#) or the [Microsoft Community Promise](#). If you would prefer a written license, or if the technologies described in this documentation are not covered by the Open Specifications Promise or Community Promise, as applicable, patent licenses are available by contacting iplg@microsoft.com.
- **License Programs.** To see all of the protocols in scope under a specific license program and the associated patents, visit the [Patent Map](#).
- **Trademarks.** The names of companies and products contained in this documentation might be covered by trademarks or similar intellectual property rights. This notice does not grant any licenses under those rights. For a list of Microsoft trademarks, visit www.microsoft.com/trademarks.
- **Fictitious Names.** The example companies, organizations, products, domain names, email addresses, logos, people, places, and events that are depicted in this documentation are fictitious. No association with any real company, organization, product, domain name, email address, logo, person, place, or event is intended or should be inferred.

Reservation of Rights. All other rights are reserved, and this notice does not grant any rights other than as specifically described above, whether by implication, estoppel, or otherwise.

Tools. The Open Specifications documentation does not require the use of Microsoft programming tools or programming environments in order for you to develop an implementation. If you have access to Microsoft programming tools and environments, you are free to take advantage of them. Certain Open Specifications documents are intended for use in conjunction with publicly available standards specifications and network programming art and, as such, assume that the reader either is familiar with the aforementioned material or has immediate access to it.

Support. For questions and support, please contact dochelp@microsoft.com.

Revision Summary

Date	Revision History	Revision Class	Comments
3/12/2010	1.0	Major	First Release.
4/23/2010	1.0.1	Editorial	Changed language and formatting in the technical content.
6/4/2010	1.0.2	Editorial	Changed language and formatting in the technical content.
7/16/2010	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
8/27/2010	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
10/8/2010	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
11/19/2010	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
1/7/2011	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
2/11/2011	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
3/25/2011	1.0.2	None	No changes to the meaning, language, or formatting of the technical content.
5/6/2011	2.0	Major	Updated and revised the technical content.
6/17/2011	3.0	Major	Updated and revised the technical content.
9/23/2011	3.0	None	No changes to the meaning, language, or formatting of the technical content.
12/16/2011	3.0	None	No changes to the meaning, language, or formatting of the technical content.
3/30/2012	3.0	None	No changes to the meaning, language, or formatting of the technical content.
7/12/2012	3.1	Minor	Clarified the meaning of the technical content.
10/25/2012	3.1	None	No changes to the meaning, language, or formatting of the technical content.
1/31/2013	3.1	None	No changes to the meaning, language, or formatting of the technical content.
8/8/2013	3.1	None	No changes to the meaning, language, or formatting of the technical content.
11/14/2013	3.1	None	No changes to the meaning, language, or formatting of the technical content.
2/13/2014	3.1	None	No changes to the meaning, language, or formatting of the technical content.
5/15/2014	3.1	None	No changes to the meaning, language, or formatting of the technical content.

Date	Revision History	Revision Class	Comments
6/30/2015	3.1	None	No changes to the meaning, language, or formatting of the technical content.
7/14/2016	3.1	None	No changes to the meaning, language, or formatting of the technical content.
6/1/2017	3.1	None	No changes to the meaning, language, or formatting of the technical content.

Table of Contents

1	Introduction	7
1.1	Glossary	7
1.2	References	8
1.2.1	Normative References	8
1.2.2	Informative References	9
1.3	Overview	9
1.4	Relationship to Other Protocols	9
1.5	Prerequisites/Preconditions	10
1.6	Applicability Statement	10
1.7	Versioning and Capability Negotiation	10
1.8	Vendor-Extensible Fields	11
1.9	Standards Assignments.....	11
2	Messages.....	12
2.1	Transport	12
2.2	Common Message Syntax	12
2.2.1	Namespaces	12
2.2.2	Messages.....	12
2.2.2.1	SignMessageRequest	13
2.2.2.2	SignMessageResponse	14
2.2.2.3	VerifyMessageRequest	14
2.2.2.4	VerifyMessageResponse	15
2.2.2.5	IssueRequest.....	15
2.2.2.6	IssueResponse.....	16
2.2.2.7	LogoutRequest.....	16
2.2.2.8	LogoutResponse	17
2.2.2.9	CreateErrorMessageRequest	18
2.2.2.10	CreateErrorMessageResponse	18
2.2.3	Elements	19
2.2.4	Complex Types.....	19
2.2.4.1	RequestType	19
2.2.4.2	ResponseType	19
2.2.4.3	PrincipalType.....	19
2.2.4.4	SamlMessageType.....	20
2.2.4.5	PostBindingType	20
2.2.4.6	RedirectBindingType.....	21
2.2.5	Simple Types	21
2.2.5.1	LogoutStatusType	21
2.2.5.2	PrincipalTypes	22
2.2.6	Attributes	22
2.2.7	Groups	22
2.2.8	Attribute Groups.....	22
3	Protocol Details	23
3.1	Common Details	23
3.1.1	Abstract Data Model.....	23
3.1.2	Timers	23
3.1.3	Initialization.....	23
3.1.4	Message Processing Events and Sequencing Rules	23
3.1.4.1	SignMessage	24
3.1.4.1.1	Messages	24
3.1.4.1.1.1	SignMessageRequest.....	24
3.1.4.1.1.2	SignMessageResponse.....	24
3.1.4.2	VerifyMessage	24
3.1.4.2.1	Messages	24

3.1.4.2.1.1	VerifyMessageRequest	24
3.1.4.2.1.2	VerifyMessageResponse	24
3.1.4.3	Issue	25
3.1.4.3.1	Messages	25
3.1.4.3.1.1	IssueRequest	25
3.1.4.3.1.2	IssueResponse	25
3.1.4.4	Logout	25
3.1.4.4.1	Messages	25
3.1.4.4.1.1	LogoutRequest	25
3.1.4.4.1.2	LogoutResponse	25
3.1.4.5	CreateErrorMessage	25
3.1.4.5.1	Messages	26
3.1.4.5.1.1	CreateErrorMessageRequest	26
3.1.4.5.1.2	CreateErrorMessageResponse	26
3.1.4.6	Types Common to Multiple Operations	26
3.1.4.6.1	Complex Types	26
3.1.4.6.1.1	PrincipalType	26
3.1.4.6.1.2	SamlMessageType	26
3.1.4.6.1.3	PostBindingType	26
3.1.4.6.1.4	RedirectBindingType	27
3.1.4.6.2	Simple Types	27
3.1.4.6.2.1	LogoutStatusType	27
3.1.4.6.2.2	PrincipalTypes	27
3.1.4.7	Status Codes for Operations	27
3.1.4.7.1	Element <Status>	27
3.1.4.7.2	Element <StatusCode>	28
3.1.4.7.3	Element <StatusMessage>	30
3.1.4.7.4	Element <StatusDetail>	30
3.1.5	Timer Events	30
3.1.6	Other Local Events	30
3.2	Server Details	30
3.2.1	Abstract Data Model	30
3.2.2	Timers	30
3.2.3	Initialization	31
3.2.4	Message Processing Events and Sequencing Rules	31
3.2.5	Timer Events	31
3.2.6	Other Local Events	31
3.3	Client Details	31
3.3.1	Abstract Data Model	31
3.3.2	Timers	31
3.3.3	Initialization	31
3.3.4	Message Processing Events and Sequencing Rules	31
3.3.5	Timer Events	31
3.3.6	Other Local Events	32
4	Protocol Examples	33
4.1	Issue Operation Examples	33
4.1.1	IssueRequest Example	33
4.1.2	IssueResponse Example	34
4.1.3	IssueResponse Example Using Artifact Binding	35
4.2	CreateErrorMessage Operation Examples	36
4.2.1	CreateErrorMessageRequest Example	36
4.2.2	CreateErrorMessageResponse Example	37
4.3	SignMessage Operation Examples	37
4.3.1	SignMessageRequest Example	37
4.3.2	SignMessageResponse Example	38
4.4	VerifyMessage Operation Examples	39
4.4.1	VerifyMessageRequest Example	39

4.4.2	VerifyMessageResponse Example.....	40
4.4.3	VerifyMessageResponse Example Using Redirect Binding	40
4.5	Logout Operations Examples	41
4.5.1	LogoutRequest Example	41
4.5.2	LogoutResponse Example	42
4.5.3	LogoutRequest Example - Locally Initiated	43
4.5.4	LogoutResponse Example:Final Response to Locally Initiated Request	43
4.5.5	LogoutRequest Example with SAMLResponse and RelayState	43
4.5.6	LogoutResponse Example with SAMLRequest and RelayState	45
5	Security	46
5.1	Security Considerations for Implementers	46
5.2	Index of Security Parameters	46
6	Appendix A: Full WSDL	47
7	Appendix B: Product Behavior	48
8	Change Tracking.....	49
9	Index.....	50

1 Introduction

This document specifies the Security Assertion Markup Language (SAML) Proxy Request Signing Protocol, which allows proxy servers to perform operations that require knowledge of configured keys and other state information about federated sites known by the **Security Token Service (STS)** server.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Active Directory Federation Services (AD FS) Proxy Server: An AD FS 2.0 service that processes SAML Federation Protocol messages. **AD FS proxy servers** are clients for the Security Assertion Markup Language (SAML) Proxy Request Signing Protocol (SAMLPR).

Active Directory Federation Services (AD FS) Security Token Service (STS): An AD FS 2.0 service that holds configuration information about federated sites. **AD FS STS** servers are servers for the Security Assertion Markup Language (SAML) Proxy Request Signing Protocol (SAMLPR).

certificate: A certificate is a collection of attributes and extensions that can be stored persistently. The set of attributes in a certificate can vary depending on the intended usage of the certificate. A certificate securely binds a public key to the entity that holds the corresponding private key. A certificate is commonly used for authentication and secure exchange of information on open networks, such as the Internet, extranets, and intranets. Certificates are digitally signed by the issuing certification authority (CA) and can be issued for a user, a computer, or a service. The most widely accepted format for certificates is defined by the ITU-T X.509 version 3 international standards. For more information about attributes and extensions, see [\[RFC3280\]](#) and [\[X509\]](#) sections 7 and 8.

SAML Artifact Binding: A method of transmitting **SAML messages** via references in HTTP messages, as specified in [\[SamIBinding\]](#) section 3.6.

SAML Identity Provider (IdP): A provider of **SAML** assertions, as specified in [\[SAMLCore2\]](#) section 2.

SAML Message: A **SAML** protocol message, as specified in [\[SAMLCore2\]](#) and [\[SamIBinding\]](#).

SAML Post Binding: A method of transmitting **SAML messages** via HTTP POST actions, as specified in [\[SamIBinding\]](#) section 3.5.

SAML Redirect Binding: A method of transmitting **SAML messages** via HTTP redirects, as specified in [\[SamIBinding\]](#) section 3.4.

SAML Service Provider (SP): A consumer of **SAML** assertions, as specified in [\[SAMLCore2\]](#) section 2.

Security Assertion Markup Language (SAML): The set of specifications that describe security assertions encoded in XML, profiles for attaching assertions to protocols and frameworks, request/response protocols used to obtain assertions, and the protocol bindings to transfer protocols, such as **SOAP** and HTTP.

security token service (STS): A web service that issues security tokens. That is, it makes assertions based on evidence that it trusts; these assertions are for consumption by whoever trusts it.

SHA-1 hash: A hashing algorithm as specified in [\[FIPS180-2\]](#) that was developed by the National Institute of Standards and Technology (NIST) and the National Security Agency (NSA).

SOAP: A lightweight protocol for exchanging structured information in a decentralized, distributed environment. **SOAP** uses XML technologies to define an extensible messaging framework, which provides a message construct that can be exchanged over a variety of underlying protocols. The framework has been designed to be independent of any particular programming model and other implementation-specific semantics. SOAP 1.2 supersedes SOAP 1.1. See [\[SOAP1.2-1/2003\]](#).

SOAP body: A container for the payload data being delivered by a **SOAP message** to its recipient. See [\[SOAP1.2-1/2007\]](#) section 5.3 for more information.

SOAP message: An XML document consisting of a mandatory SOAP envelope, an optional SOAP header, and a mandatory **SOAP body**. See [\[SOAP1.2-1/2007\]](#) section 5 for more information.

Uniform Resource Locator (URL): A string of characters in a standardized format that identifies a document or resource on the World Wide Web. The format is as specified in [\[RFC1738\]](#).

Web Services Description Language (WSDL): An XML format for describing network services as a set of endpoints that operate on messages that contain either document-oriented or procedure-oriented information. The operations and messages are described abstractly and are bound to a concrete network protocol and message format in order to define an endpoint. Related concrete endpoints are combined into abstract endpoints, which describe a network service. WSDL is extensible, which allows the description of endpoints and their messages regardless of the message formats or network protocols that are used.

XML namespace: A collection of names that is used to identify elements, types, and attributes in XML documents identified in a URI reference [\[RFC3986\]](#). A combination of XML namespace and local name allows XML documents to use elements, types, and attributes that have the same names but come from different sources. For more information, see [\[XMLNS-2ED\]](#).

XML Schema (XSD): A language that defines the elements, attributes, namespaces, and data types for XML documents as defined by [\[XMLSCHEMA1/2\]](#) and [\[XMLSCHEMA2/2\]](#) standards. An XML schema uses XML syntax for its language.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[SamlBinding] Cantor, S., Hirsch, F., Kemp, J., et al., "Bindings for the OASIS Security Assertion Markup Language (SAML) V2.0", March 2005, <http://docs.oasis-open.org/security/saml/v2.0/saml-bindings-2.0-os.pdf>

[SAMLCore2] Cantor, S., Kemp, J., Philpott, R., and Maler, E., Eds., "Assertions and Protocol for the OASIS Security Assertion Markup Language (SAML) V2.0", March 2005, <http://docs.oasis-open.org/security/saml/v2.0/saml-core-2.0-os.pdf>

[SOAP1.2-1/2003] Gudgin, M., Hadley, M., Mendelsohn, N., et al., "SOAP Version 1.2 Part 1: Messaging Framework", W3C Recommendation, June 2003, <http://www.w3.org/TR/2003/REC-soap12-part1-20030624>

[WSAddressing] Box, D., et al., "Web Services Addressing (WS-Addressing)", August 2004, <http://www.w3.org/Submission/ws-addressing/>

[WSDL] Christensen, E., Curbera, F., Meredith, G., and Weerawarana, S., "Web Services Description Language (WSDL) 1.1", W3C Note, March 2001, <https://www.w3.org/TR/2001/NOTE-wsdl-20010315>

[WSSC1.3] Lawrence, K., Kaler, C., Nadalin, A., et al., "WS-SecureConversation 1.3", March 2007, <http://docs.oasis-open.org/ws-sx/ws-secureconversation/200512/ws-secureconversation-1.3-os.html>

[WSSU1.0] OASIS Standard, "WS Security Utility 1.0", 2004, <http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-utility-1.0.xsd>

[WSTrust] IBM, Microsoft, Nortel, VeriSign, "WS-Trust V1.0", February 2005, <http://specs.xmlsoap.org/ws/2005/02/trust/WS-Trust.pdf>

[XMLNS] Bray, T., Hollander, D., Layman, A., et al., Eds., "Namespaces in XML 1.0 (Third Edition)", W3C Recommendation, December 2009, <https://www.w3.org/TR/2009/REC-xml-names-20091208/>

[XMLSCHEMA1] Thompson, H., Beech, D., Maloney, M., and Mendelsohn, N., Eds., "XML Schema Part 1: Structures", W3C Recommendation, May 2001, <https://www.w3.org/TR/2001/REC-xmlschema-1-20010502/>

[XMLSCHEMA2] Biron, P.V., Ed. and Malhotra, A., Ed., "XML Schema Part 2: Datatypes", W3C Recommendation, May 2001, <https://www.w3.org/TR/2001/REC-xmlschema-2-20010502/>

1.2.2 Informative References

[WS-Trust1.3] Nadalin, A., Goodner, M., Gudgin, M., Barbir, A., Granqvist, H., "WS-Trust 1.3", OASIS Standard 19 March 2007, <http://docs.oasis-open.org/ws-sx/ws-trust/200512/ws-trust-1.3-os.html>

1.3 Overview

The Security Assertion Markup Language (SAML) Proxy Request Signing Protocol (SAMLPR) provides the capability for **AD FS proxy servers** to have the **AD FS STS** server for an installation perform operations that require knowledge of the configured keys and other state information about federated sites known by the **Security Token Service (STS)** server. For more information, see [\[WS-Trust1.3\]](#). In particular, proxy servers use the SAMLPR Protocol to have the STS server in an installation perform **SAML** (see [\[SAMLCore2\]](#) and [\[SamlBinding\]](#)) signature operations upon messages to be sent. Multiple proxy servers can use a single STS server.

The protocol is stateless, with the parameters of each message being fully self-contained.

1.4 Relationship to Other Protocols

The Security Assertion Markup Language (SAML) Proxy Request Signing Protocol (SAMLPR) uses **SOAP** over TCP for local connections, as shown in the following layering diagram:

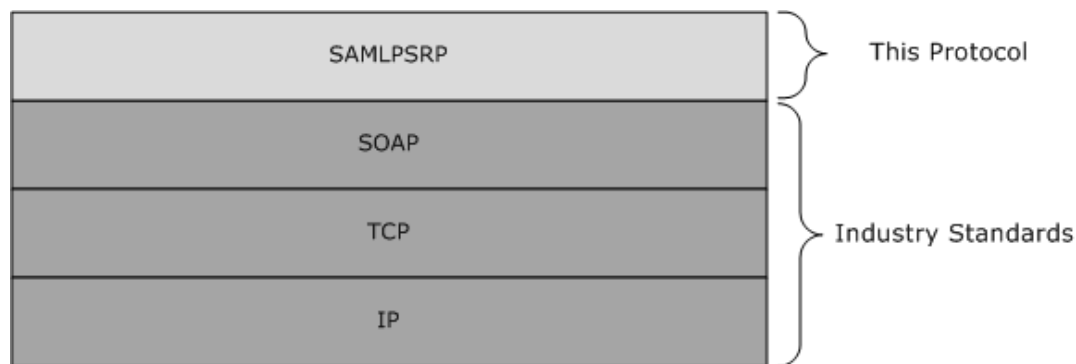


Figure 1: SAMLPR SOAP over TCP layer diagram

The Security Assertion Markup Language (SAML) Proxy Request Signing Protocol (SAMLPR) uses SOAP over HTTPS for remote connections, as shown in the following layering diagram:

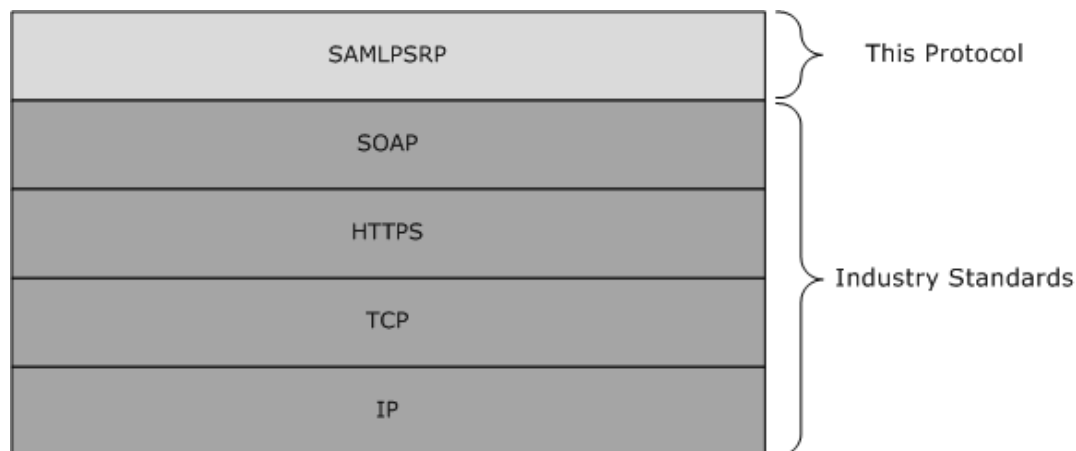


Figure 2: SAMLPR SOAP over HTTPS layer diagram

1.5 Prerequisites/Preconditions

The client is configured with the **Uniform Resource Locator (URL)** of the server's **SOAP** service in order to call the service.

1.6 Applicability Statement

The SAMLPR Protocol is used by services that perform **SAML** signature operations for proxy servers by **STS** servers in a manner that is compatible with AD FS 2.0.

1.7 Versioning and Capability Negotiation

This protocol uses the versioning mechanisms defined in the following specification:

- **SOAP** 1.2, as specified in [\[SOAP1.2-1/2003\]](#).

This protocol does not perform any capability negotiation.

1.8 Vendor-Extensible Fields

The schema for this protocol provides for extensibility points for additional elements to be added to each **SOAP message** body. Elements within these extensibility points that are not understood are ignored.

1.9 Standards Assignments

There are no standards assignments for this protocol beyond those defined in the following specification:

- **SOAP** 1.2, as specified in [\[SOAP1.2-1/2003\]](#).

2 Messages

2.1 Transport

The Security Assertion Markup Language (SAML) Proxy Request Signing Protocol uses **SOAP**, as specified in [\[SOAP1.2-1/2003\]](#), over TCP locally or HTTPS remotely, for communication.

2.2 Common Message Syntax

This section contains no common definitions used by this protocol.

2.2.1 Namespaces

This specification defines and references various **XML namespaces** using the mechanisms specified in [\[XMLNS\]](#). Although this specification associates a specific XML namespace prefix for each XML namespace that is used, the choice of any particular XML namespace prefix is implementation-specific and not significant for interoperability.

Prefix	Namespace URI	Reference
s	http://www.w3.org/2003/05/soap-envelope	[SOAP1.2-1/2003]
xs	http://www.w3.org/2001/XMLSchema	[XMLSCHEMA1] and [XMLSCHEMA2]
a	http://schemas.xmlsoap.org/ws/2004/08/addressing	[WSAddressing] section 1.2
msis	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol	This document ([MS-SAMLPR])
samlp	urn:oasis:names:tc:SAML:2.0:protocol	[SAMLCore2]
saml	urn:oasis:names:tc:SAML:2.0:assertion	[SAMLCore2]
wst	http://docs.oasis-open.org/ws-sx/ws-trust/200512	[WSTrust]
wssc	http://docs.oasis-open.org/ws-sx/ws-secureconversation/200512	[WSSC1.3]
wssu	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-utility-1.0.xsd	[WSSU1.0]

2.2.2 Messages

Message	Description
SignMessageRequest	A message that requests that a SAML Message signature be applied to a SAML Message, if the configuration for the requested principal specifies that messages are to be signed.
SignMessageResponse	A reply message to SignMessageRequest, containing the resulting SAML Message, which is signed, if the configuration for the requested principal specifies that messages are to be signed.
VerifyMessageRequest	A message that requests verification that a SAML Message is from a known party and signed according to the metadata directives for that party.

Message	Description
VerifyMessageResponse	A reply message to the VerifyMessageRequest message, containing a Boolean result.
IssueRequest	A message requesting issuance of a SAML token.
IssueResponse	A reply message to the IssueRequest message containing a SAML response message.
LogoutRequest	A message requesting that a SAML logout be performed.
LogoutResponse	A reply message to the LogoutRequest message containing updated SessionState and LogoutState values.
CreateErrorMessageRequest	A message that requests creation of a SAML error message, which will be signed, if the configuration for the requested principal specifies that messages are to be signed.
CreateErrorMessageResponse	A reply message to the CreateErrorMessageRequest message containing the created SAML error message.

2.2.2.1 SignMessageRequest

The SignMessageRequest message requests that a **SAML Message** signature be applied to a SAML Message, if the configuration for the requested principal specifies that messages are to be signed. It is used by the following message:

Message type	Action URI
Request	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest

body: The **SOAP body** MUST contain a single msis:SignMessageRequest element with the following type:

```

<complexType name="SignMessageRequestType">
  <complexContent>
    <extension base="msis:RequestType">
      <sequence>
        <element name="ActivityId" type="string"/>
        <element name="Message" type="msis:SamlMessageType"/>
        <element name="Principal" type="msis:PrincipalType"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>

```

ActivityId: An opaque string supplied by the caller to track the activity to which this message pertains.

Message: A complex type representing a **SAML** Protocol message.

Principal: A complex type representing a SAML EntityId for a **SAML Identity Provider (IdP)**, a **SAML Service Provider (SP)**, or this **STS** server.

2.2.2.2 SignMessageResponse

A SignMessageResponse message is a reply message to SignMessageRequest, containing the resulting **SAML Message**, which is signed, if the configuration for the requested principal specifies that messages are to be signed. It is used by the following message:

Message type	Action URI
Response	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse

body: The **SOAP body** MUST contain a single msis:SignMessageResponse element with the following type:

```
<complexType name="SignMessageResponseType">
  <complexContent>
    <extension base="msis:ResponseType">
      <sequence>
        <element name="Message" type="msis:SamlMessageType"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>
```

Message: A complex type representing a **SAML** Protocol message.

2.2.2.3 VerifyMessageRequest

The VerifyMessageRequest message requests verification that a **SAML Message** is from a known party and signed according to the metadata directives for that party. It is used by the following message:

Message type	Action URI
Request	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest

body: The **SOAP body** MUST contain a single msis:VerifyMessageRequest element with the following type:

```
<complexType name="VerifyMessageRequestType" >
  <complexContent>
    <extension base="msis:RequestType">
      <sequence>
        <element name="ActivityId" type="string"/>
        <element name="Message" type="msis:SamlMessageType"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>
```

ActivityId: An opaque string supplied by the caller to track the activity to which this message pertains.

Message: A complex type representing a **SAML** Protocol message.

2.2.2.4 VerifyMessageResponse

The VerifyMessageResponse message is a reply to VerifyMessageRequest, containing a Boolean result. It is used by the following message:

Message type	Action URI
Response	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse

body: The **SOAP body** MUST contain a single msis:VerifyMessageResponse element with the following type:

```
<complexType name="VerifyMessageResponseType" >
  <complexContent>
    <extension base="msis:ResponseType">
      <sequence>
        <element name="IsVerified" type="boolean"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
```

IsVerified: A Boolean result indicating whether a **SAML Message** is from a known party and signed according to the metadata directives for that party.

2.2.2.5 IssueRequest

The IssueRequest message requests the issuance of a **SAML** token. It is used by the following message:

Message type	Action URI
Request	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest

body: The **SOAP body** MUST contain a single msis:IssueRequest element with the following type:

```
<complexType name="IssueRequestType" >
  <complexContent>
    <extension base="msis:RequestType">
      <sequence>
        <element name="ActivityId" type="string"/>
        <element name="Message" type="msis:SamlMessageType"/>
        <element name="OnBehalfOf" type="wst:OnBehalfOfType"/>
        <element name="SessionState" type="string"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
/>
      </sequence>
    </extension>
  </complexContent>
</complexType>
```

ActivityId: An opaque string supplied by the caller to track the activity to which this message pertains.

Message: A complex type representing a SAML Protocol message.

OnBehalfOf: A complex type representing the party to issue the token for.

SessionState: A structured string representing the information required to log out from this session.

2.2.2.6 IssueResponse

The IssueResponse message is a reply to IssueRequest, containing a **SAML** response message. It is used by the following message:

Message type	Action URI
Response	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse

body: The **SOAP body** MUST contain a single msis:IssueResponse element with the following type:

```
<complexType name="IssueResponseType">
  <complexContent>
    <extension base="msis:ResponseType">
      <sequence>
        <element name="Message" minOccurs="0" type="msis:SamlMessageType"/>
        <element name="SessionState" type="string"/>
        <element name="AuthenticatingProvider" type="string"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>
```

Message: A complex type representing a SAML Protocol message.

SessionState: A structured string representing the information required to log out from this session.

AuthenticatingProvider: The URI of a claims provider or a local **STS** identifier, depending upon where the user authenticated.

2.2.2.7 LogoutRequest

The LogoutRequest message requests that a **SAML** logout be performed. It is used by the following message:

Message type	Action URI
Request	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest

body: The **SOAP body** MUST contain a single msis:LogoutRequest element with the following type:

```
<complexType name="LogoutRequestType" >
  <complexContent>
    <extension base="msis:RequestType">
```



```

        <sequence>
          <element name="ActivityId" type="string"/>
          <element name="Message" minOccurs="0" type="msis:SamlMessageType"/>
          <element name="SessionState" type="string"/>
          <element name="LogoutState" type="string"/>
          <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
        />
      </sequence>
    </extension>
  </complexContent>
</complexType>

```

ActivityId: An opaque string supplied by the caller to track the activity that this message pertains to.

Message: A complex type representing a SAML protocol message.

SessionState: A structured string representing the information required to log out from this session.

LogoutState: A structured string representing additional information required to log out from this session.

2.2.2.8 LogoutResponse

The LogoutResponse message is a reply to LogoutRequest, containing updated SessionState and LogoutState values. It is used by the following message:

Message type	Action URI
Response	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse

body: The **SOAP body** MUST contain a single msis:LogoutResponse element with the following type:

```

<complexType name="LogoutResponseType">
  <complexContent>
    <extension base="msis:ResponseType">
      <sequence>
        <element name="LogoutStatus" type="msis:LogoutStatusType"/>
        <element name="Message" type="msis:SamlMessageType" minOccurs="0"/>
        <element name="SessionState" type="string"/>
        <element name="LogoutState" type="string"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>

```

LogoutStatus: A complex type representing the status of the logout process.

Message: A complex type representing a **SAML** Protocol message.

SessionState: A structured string representing the information required to log out from this session.

LogoutState: A structured string representing additional information required to log out from this session.

2.2.2.9 CreateErrorMessageRequest

The CreateErrorMessageRequest message requests the creation of a **SAML** error message, which will be signed, if the configuration for the requested principal specifies that messages are to be signed. It is used by the following message:

Message type	Action URI
Request	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest

body: The **SOAP body** MUST contain a single msis:CreateErrorMessageRequest element with the following type:

```
<complexType name="CreateErrorMessageRequestType">
  <complexContent>
    <extension base="msis:RequestType">
      <sequence>
        <element name="ActivityId" type="string"/>
        <element name="Message" type="msis:SamlMessageType"/>
        <element name="Principal" type="msis:PrincipalType"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
</complexType>
```

ActivityId: An opaque string supplied by the caller to track the activity to which this message pertains.

Message: A complex type representing a SAML Protocol message.

Principal: A complex type representing a SAML EntityId for a **SAML IdP**, a **SAML SP**, or this **STS** server.

2.2.2.10 CreateErrorMessageResponse

The CreateErrorMessageResponse message is a reply to CreateErrorMessageRequest, containing the created **SAML** error message. It is used by the following messages:

Message type	Action URI
Response	http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse

body: The **SOAP body** MUST contain a single msis:CreateErrorMessageResponse element with the following type:

```
<complexType name="CreateErrorMessageResponseType">
  <complexContent>
    <extension base="msis:ResponseType">
      <sequence>
        <element name="Message" type="msis:SamlMessageType"/>
        <any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"
      />
    </sequence>
  </extension>
</complexContent>
```

</complexType>

Message: A complex type representing a SAML Protocol message.

2.2.3 Elements

This specification does not define any common **XML Schema** element definitions.

2.2.4 Complex Types

The following table summarizes the set of common XML schema complex type definitions defined by this specification. XML schema complex type definitions that are specific to a particular operation are described with the operation.

Complex type	Description
RequestType	An abstract type containing protocol request message parameters.
ResponseType	An abstract type containing protocol response messages parameters.
PrincipalType	A structure containing a PrincipalTypes value and an identifier for the principal.
SamlMessageType	A structure containing a representation of a SAML Protocol message.
PostBindingType	A structure containing SAML binding information for a SAML post binding .
RedirectBindingType	A structure containing SAML binding information for a SAML redirect binding .

2.2.4.1 RequestType

This abstract type contains request message parameters for messages using this protocol. The schema for this type MUST be as follows:

```
<complexType name="RequestType" abstract="true"/>
```

2.2.4.2 ResponseType

This abstract type contains response message parameters for messages using this protocol. The schema for this type MUST be as follows:

```
<complexType name="ResponseType" abstract="true"/>
```

2.2.4.3 PrincipalType

This structure contains a PrincipalTypes value and an identifier for the principal. The schema for this type MUST be as follows:

```
<complexType name="PrincipalType">  
  <sequence>
```

```

    <element name="Type" type="msis:PrincipalTypes"/>
    <element name="Identifier" type="string"/>
  </sequence>
</complexType>

```

Type: A PrincipalTypes enumeration value identifying the type of the **SAML** principal.

Identifier: An identifier for the SAML principal. This is a SAML EntityId.

2.2.4.4 SamlMessageType

This structure contains a representation of a **SAML** Protocol message. The schema for this type MUST be as follows:

```

<complexType name="SamlMessageType">
  <sequence>
    <element name="BaseUri" type="anyURI"/>
    <choice>
      <element name="SAMLart" type="string"/>
      <element name="SAMLRequest" type="string"/>
      <element name="SAMLResponse" type="string"/>
    </choice>
    <choice>
      <element name="PostBindingInformation" type="msis:PostBindingType"/>
      <element name="RedirectBindingInformation" type="msis:RedirectBindingType"/>
    </choice>
  </sequence>
</complexType>

```

BaseUri: The **URL** to post message to.

SAMLart: A SAML artifact identifier, base64-encoded as per [\[SamlBinding\]](#) section 3.6.

SAMLRequest: A SAML request message, base64-encoded as per [\[SamlBinding\]](#) sections 3.4 and 3.5.

SAMLResponse: A SAML response message, base64-encoded as per [\[SamlBinding\]](#) sections 3.4 and 3.5.

PostBindingInformation: Information about the **SAML Message** using the **SAML post binding**, as per [\[SamlBinding\]](#) section 3.5.

RedirectBindingInformation: Information about the SAML Message using the **SAML redirect binding**, as per [\[SamlBinding\]](#) section 3.4.

2.2.4.5 PostBindingType

This structure contains **SAML** binding information for a **SAML post binding**. The schema for this type MUST be as follows:

```

<complexType name="PostBindingType">
  <sequence>
    <element name="RelayState" minOccurs="0" type="string"/>
  </sequence>
</complexType>

```

RelayState: An opaque BLOB that, if present in the request, MUST be returned in the response, as per [\[SamlBinding\]](#) section 3.5.3.

2.2.4.6 RedirectBindingType

This structure contains **SAML** binding information for a **SAML redirect binding**. The schema for this type MUST be as follows:

```
<complexType name="RedirectBindingType">
  <sequence>
    <element name="RelayState" minOccurs="0" type="string"/>
    <sequence minOccurs="0">
      <element name="Signature" type="string"/>
      <element name="SigAlg" type="string"/>
      <element name="QueryStringHash" minOccurs="0" type="string"/>
    </sequence>
  </sequence>
</complexType>
```

RelayState: An opaque BLOB that, if present in the request, MUST be returned in the response, as per [\[SamlBinding\]](#) section 3.4.3.

Signature: The message signature (if present), encoded as per [\[SamlBinding\]](#) section 3.4.4.1.

SigAlg: The message signature algorithm (if present), as per [\[SamlBinding\]](#) section 3.4.4.1.

QueryStringHash: A base64-encoded **SHA-1 hash** of the redirect query string (if present), for integrity purposes, as per [\[SamlBinding\]](#) section 3.6.4.

2.2.5 Simple Types

The following table summarizes the set of common XML schema simple type definitions defined by this specification. XML schema simple type definitions that are specific to a particular operation are described with the operation.

Simple type	Description
LogoutStatusType	An enumeration of status values for logout operations.
PrincipalTypes	An enumeration of the types of SAML principals.

2.2.5.1 LogoutStatusType

This type enumerates the set of status values for logout operations. The schema for this type MUST be as follows:

```
<simpleType name="LogoutStatusType">
  <restriction base="string">
    <enumeration value="InProgress" />
    <enumeration value="LogoutPartial" />
    <enumeration value="LogoutSuccess" />
  </restriction>
</simpleType>
```

InProgress: Indicates that more logout work is required to be performed.

LogoutPartial: Indicates that the logout process is complete, but all session participants might not have been logged out.

LogoutSuccess: Indicates the logout process is complete, with all session participants logged out.

2.2.5.2 PrincipalTypes

This type enumerates the set of types of **SAML** principals. The schema for this type **MUST** be as follows:

```
<simpleType name="PrincipalTypes">
  <restriction base="string">
    <enumeration value="Self" />
    <enumeration value="Scope" />
    <enumeration value="Authority" />
  </restriction>
</simpleType>
```

Self: Indicates that the principal is this **STS** server.

Scope: Indicates that the principal is a **SAML Service Provider**, identified by an Entity Identifier, as per [\[SAMLCore2\]](#) section 8.3.6.

Authority: Indicates that the principal is a **SAML Identity Provider**, identified by an Entity Identifier, as per [\[SAMLCore2\]](#) section 8.3.6.

2.2.6 Attributes

This specification does not define any common XML schema attribute definitions.

2.2.7 Groups

This specification does not define any common XML schema group definitions.

2.2.8 Attribute Groups

This specification does not define any common XML schema attribute group definitions.

3 Protocol Details

3.1 Common Details

This section describes protocol details that are common among multiple port types.

3.1.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

The SAMLPR Protocol enables proxy servers to have **STS** servers perform operations requiring state held at the STS server. Other than standard **SOAP** request/response protocol state that is not specific to this protocol, no state about the protocol is maintained at either the protocol client or server.

3.1.2 Timers

There are no protocol-specific timer events that **MUST** be serviced by an implementation. This protocol does not require timers beyond those that are used by the underlying transport to transmit and receive **SOAP messages**. The protocol does not include provisions for time-based retry for sending protocol messages.

3.1.3 Initialization

No protocol-specific initialization is required to use this protocol. Standard **SOAP** bindings **MUST** be established between the client and server before initiating communication.

For clients running on the local machine, the standard **STS** server SOAP endpoint address is `net.tcp://localhost/samlprotocol`. For clients running on remote machines connecting to a server, the standard STS server SOAP endpoint address is `https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the server domain name. Other port addresses **MAY** be used by implementations. [<1>](#)

3.1.4 Message Processing Events and Sequencing Rules

The following table summarizes the list of operations as defined by this specification:

Operation	Description
SignMessage	This operation causes a SAML Message signature be applied to the supplied SAML Message when the configuration requires signing, with the resulting message being returned as a result.
VerifyMessage	This operation verifies whether a SAML Message is from a known party and signed according to metadata directives for that party, returning the result as a Boolean.
Issue	This operation causes issuance of a SAML token.
Logout	This operation causes a SAML session to be logged out.
CreateErrorMessage	This operation creates a SAML error message, applying a signature, if the configuration for the requested principal specifies that messages are to be signed.

For each operation there is a request and reply message. In all cases, the sequence of operation is that the client sends the request message to the server, which responds with the corresponding reply message. The server **MUST** accept the request messages and the client **MUST** accept the corresponding reply messages, when sent in response to a request message. The behavior of any other uses of these messages is undefined.

3.1.4.1 SignMessage

This operation causes a **SAML Message** signature be applied to the supplied SAML Message when the configuration requires signing, with the resulting message being returned as a result. This operation consists of the client sending a SignMessageRequest message to the server, which replies with a SignMessageResponse message.

3.1.4.1.1 Messages

The following table summarizes the set of message definitions that are specific to this operation.

Message	Description
SignMessageRequest	Conveys request parameters for SignMessage operation.
SignMessageResponse	Conveys response parameters for SignMessage operation.

3.1.4.1.1.1 SignMessageRequest

This message conveys request parameters for the SignMessage operation.

3.1.4.1.1.2 SignMessageResponse

This message conveys response parameters for the SignMessage operation.

3.1.4.2 VerifyMessage

This operation verifies whether a **SAML Message** is from a known party and signed according to metadata directives for that party, returning the result as a Boolean. This operation consists of the client sending a VerifyMessageRequest message to the server, which replies with a VerifyMessageResponse message.

3.1.4.2.1 Messages

The following table summarizes the set of message definitions that are specific to this operation.

Message	Description
VerifyMessageRequest	Conveys request parameters for the VerifyMessage operation.
VerifyMessageResponse	Conveys response parameters for the VerifyMessage operation.

3.1.4.2.1.1 VerifyMessageRequest

This message conveys request parameters for the VerifyMessage operation.

3.1.4.2.1.2 VerifyMessageResponse

This message conveys response parameters for the VerifyMessage operation.

3.1.4.3 Issue

This operation causes the issuance of a **SAML** token. This operation consists of the client sending an IssueRequest message to the server, which replies with an IssueResponse message.

3.1.4.3.1 Messages

The following table summarizes the set of message definitions that are specific to this operation.

Message	Description
IssueRequest	Conveys request parameters for the Issue operation.
IssueResponse	Conveys response parameters for the Issue operation.

3.1.4.3.1.1 IssueRequest

This message conveys request parameters for the Issue operation.

3.1.4.3.1.2 IssueResponse

This message conveys response parameters for the Issue operation.

3.1.4.4 Logout

This operation causes a **SAML** session to be logged out. This operation consists of the client sending a LogoutRequest message to the server, which replies with a LogoutResponse message.

3.1.4.4.1 Messages

The following table summarizes the set of message definitions that are specific to this operation.

Message	Description
LogoutRequest	Conveys request parameters for the Logout operation.
LogoutResponse	Conveys response parameters for the Logout operation.

3.1.4.4.1.1 LogoutRequest

This message conveys request parameters for the Logout operation.

3.1.4.4.1.2 LogoutResponse

This message conveys response parameters for Logout operation.

3.1.4.5 CreateErrorMessage

This operation creates a **SAML** error message, applying a signature, if the configuration for the requested principal specifies that messages are to be signed. This operation consists of the client

sending a CreateErrorMessageRequest message to the server, which replies with a CreateErrorMessageResponse message.

3.1.4.5.1 Messages

The following table summarizes the set of message definitions that are specific to this operation.

Message	Description
CreateErrorMessageRequest	Conveys request parameters for the CreateErrorMessage operation.
CreateErrorMessageResponse	Conveys response parameters for the CreateErrorMessage operation.

3.1.4.5.1.1 CreateErrorMessageRequest

This message conveys request parameters for the CreateErrorMessage operation.

3.1.4.5.1.2 CreateErrorMessageResponse

This message conveys response parameters for the CreateErrorMessage operation.

3.1.4.6 Types Common to Multiple Operations

This section describes types that are common to multiple operations.

3.1.4.6.1 Complex Types

The following table summarizes the XML schema complex type definitions that are common to multiple operations, the schemas for which are defined in section [2.2.4](#).

Complex type	Description
PrincipalType	Identifies participant in a SAML federation, including its role.
SamIMessageType	Representation of a SAML Protocol message and the binding used to send it.
PostBindingType	Information about a SAML post binding , which consists of its RelayState, if present.
RedirectBindingType	Information about a SAML redirect binding , which consists of its RelayState, if present, and signature information, if present.

3.1.4.6.1.1 PrincipalType

This complex type identifies participant in a **SAML** federation, including its role.

3.1.4.6.1.2 SamIMessageType

This complex type specifies the representation of a **SAML** Protocol message and the binding used to send it.

3.1.4.6.1.3 PostBindingType

This complex type specifies information about a **SAML post binding**, which consists of its RelayState, if present.

3.1.4.6.1.4 RedirectBindingType

This complex type specifies information about a **SAML redirect binding**, which consists of its RelayState, if present, and signature information, if present.

3.1.4.6.2 Simple Types

The following table summarizes the XML schema simple definitions that are common to multiple operations, the schemas for which are defined in section [2.2.5](#).

Simple type	Description
LogoutStatusType	Indicates whether logout operation has completed or not, and if completed, whether all session participants were logged out.
PrincipalTypes	Identifies role of participant in SAML federation.

3.1.4.6.2.1 LogoutStatusType

This simple type indicates whether logout operation has completed or not, and if completed, whether all session participants were logged out.

3.1.4.6.2.2 PrincipalTypes

This simple type identifies the role of the participant in a **SAML** federation.

3.1.4.7 Status Codes for Operations

This section describes both the <Status> element and the different status codes as specified in [\[SAMLCore2\]](#), section 3.2.2.

3.1.4.7.1 Element <Status>

The <Status> element contains the following three elements:

Element	Required/Optional	Description
<StatusCode>	Required	This element MUST contain a code that represents the status of a request that has been received by the server.
<StatusMessage>	Optional	This element MAY contain a message that is to be returned to the operator.
<StatusDetail>	Optional	This element MAY contain additional information concerning an error condition.

The following schema fragment defines both the <Status> element and its corresponding **StatusType** complex type:

```
<element name="Status" type="samlp:StatusType"/>
<complexType name="StatusType">
  <sequence>
    <element ref="samlp:StatusCode"/>
    <element ref="samlp:StatusMessage" minOccurs="0"/>
    <element ref="samlp:StatusDetail" minOccurs="0"/>
  </sequence>
```

</complexType>

3.1.4.7.2 Element <StatusCode>

The <StatusCode> element contains a code or a set of nested codes that represent the status of the request. Every <StatusCode> element has the following attribute:

Attribute	Required/Optional	Description
Value	Required	The status code value. This value MUST contain a URI reference. The Value attribute of the top-level <StatusCode> element MUST be one of the top-level status codes given in this section. Subordinate <StatusCode> elements MAY use second-level status code values given in this section.

The <StatusCode> element MAY contain subordinate second-level <StatusCode> elements that provide additional information on the error condition.

The permissible top-level status codes are:

Status code	Description
urn:oasis:names:tc:SAML:2.0:status:Success	The request succeeded.
urn:oasis:names:tc:SAML:2.0:status:Requester	The request could not be performed due to an error on the part of the requester.
urn:oasis:names:tc:SAML:2.0:status:Responder	The request could not be performed due to an error on the part of the SAML responder or SAML authority.
urn:oasis:names:tc:SAML:2.0:status:VersionMismatch	The SAML responder could not process the request because the version of the request message was incorrect.

The second-level status codes are:

Status code	Description
urn:oasis:names:tc:SAML:2.0:status:AuthnFailed	The responding provider was unable to successfully authenticate the principal.
urn:oasis:names:tc:SAML:2.0:status:InvalidAttrNameOrValue	Unexpected or invalid content was encountered within a <saml:Attribute> or <saml:AttributeValue> element.
urn:oasis:names:tc:SAML:2.0:status:InvalidNameIDPolicy	The responding provider cannot or will not support the requested name identifier policy.
urn:oasis:names:tc:SAML:2.0:status:NoAuthnContext	The specified authentication context requirements cannot be met by the responder.
urn:oasis:names:tc:SAML:2.0:status:NoAvailableIDP	Used by an intermediary to indicate that none of the supported identity provider <Loc> elements in an <IDPList> can be resolved or that none of the supported identity providers are available.
urn:oasis:names:tc:SAML:2.0:status:NoPassive	Indicates that the responding provider cannot authenticate the principal passively, as has been requested.

Status code	Description
urn:oasis:names:tc:SAML:2.0:status:NoSupportedIDP	Used by an intermediary to indicate that none of the identity providers in an <IDPList> are supported by the intermediary.
urn:oasis:names:tc:SAML:2.0:status:PartialLogout	Used by a session authority to indicate to a session participant that it was not able to propagate the logout request to all other session participants.
urn:oasis:names:tc:SAML:2.0:status:ProxyCountExceeded	Indicates that a responding provider cannot authenticate the principal directly and is not permitted to proxy the request further.
urn:oasis:names:tc:SAML:2.0:status:RequestDenied	The SAML responder or SAML authority is able to process the request but has chosen not to respond. This status code MAY be used when there is concern about the security context of the request message or the sequence of request messages received from a particular requester.
urn:oasis:names:tc:SAML:2.0:status:RequestUnsupported	The SAML responder or SAML authority does not support the request.
urn:oasis:names:tc:SAML:2.0:status:RequestVersionDeprecated	The SAML responder cannot process any requests with the protocol version specified in the request.
urn:oasis:names:tc:SAML:2.0:status:RequestVersionTooHigh	The SAML responder cannot process the request because the protocol version specified in the request message is a major upgrade from the highest protocol version supported by the responder.
urn:oasis:names:tc:SAML:2.0:status:RequestVersionTooLow	The SAML responder cannot process the request because the protocol version specified in the request message is too low.
urn:oasis:names:tc:SAML:2.0:status:ResourceNotRecognized	The resource value provided in the request message is invalid or unrecognized.
urn:oasis:names:tc:SAML:2.0:status:TooManyResponses	The response message would contain more elements than the SAML responder is able to return.
urn:oasis:names:tc:SAML:2.0:status:UnknownAttrProfile	An entity that has no knowledge of a particular attribute profile has been presented with an attribute drawn from that profile.
urn:oasis:names:tc:SAML:2.0:status:UnknownPrincipal	The responding provider does not recognize the principal specified or implied by the request.
urn:oasis:names:tc:SAML:2.0:status:UnsupportedBinding	The SAML responder cannot properly fulfill the request using the protocol binding specified in the request.

The following schema fragment defines the <StatusCode> element and its corresponding **StatusCodeType** complex type:

```
<element name="StatusCode" type="samlp:StatusCodeType"/>
<complexType name="StatusCodeType">
```

```

<sequence>
  <element ref="samlp:StatusCode" minOccurs="0"/>
</sequence>
<attribute name="Value" type="anyURI" use="required"/>
</complexType>

```

3.1.4.7.3 Element <StatusMessage>

The <StatusMessage> element specifies a message that MAY be returned to an operator. The following schema fragment defines the <StatusMessage> element:

```

<element name="StatusMessage" type="string"/>

```

3.1.4.7.4 Element <StatusDetail>

The <StatusDetail> element MAY be used to specify additional information concerning the status of the request. The additional information consists of zero or more elements from any namespace, with no requirement for a schema to be present or for schema validation of the <StatusDetail> contents.

The following schema fragment defines the <StatusDetail> element and its corresponding **StatusDetailType** complex type:

```

<element name="StatusDetail" type="samlp:StatusDetailType"/>
<complexType name="StatusDetailType">
  <sequence>
    <any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </sequence>
</complexType>

```

3.1.5 Timer Events

This protocol does not require timers beyond those that are used by the underlying transport to transmit and receive **SOAP messages**. The protocol does not include provisions for time-based retry for sending protocol messages.

3.1.6 Other Local Events

This protocol does not have dependencies on any transport protocols other than HTTP 1.1 and TCP. This protocol relies on these transport mechanisms for the correct and timely delivery of protocol messages. The protocol does not take action in response to any changes or failure in machine state or network communications.

3.2 Server Details

3.2.1 Abstract Data Model

This port type utilizes the common abstract data model described in section [3.1.1](#).

3.2.2 Timers

This port type utilizes the common timers design described in section [3.1.2](#).

3.2.3 Initialization

This port type utilizes the common initialization design described in section [3.1.3](#). In addition, an implementation SHOULD publish a **SOAP** endpoint at the port `net.tcp://localhost/samlprotocol` to be connected to by local clients. Also, an implementation SHOULD publish a SOAP endpoint at the port `https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the server domain name, to be connected to by remote clients. Other port addresses MAY be used by implementations. [<2>](#)

3.2.4 Message Processing Events and Sequencing Rules

This port type utilizes the common message processing events and sequencing rules described in section [3.1.4](#).

3.2.5 Timer Events

This port type utilizes the common timer events design described in section [3.1.5](#).

3.2.6 Other Local Events

This port type utilizes the common other local events design described in section [3.1.6](#).

3.3 Client Details

The client side of this protocol is simply a pass-through. That is, no additional timers or other state is required on the client side of this protocol. Calls made by the higher-layer protocol or implementation are passed directly to the transport, and the results returned by the transport are passed directly back to the higher-layer protocol or application.

3.3.1 Abstract Data Model

This port type utilizes the common abstract data model described in section [3.1.1](#).

3.3.2 Timers

This port type utilizes the common timers design described in section [3.1.2](#).

3.3.3 Initialization

This port type utilizes the common initialization design described in section [3.1.3](#). In addition, an implementation SHOULD connect to a **SOAP** endpoint at the port `net.tcp://localhost/samlprotocol` for a local connection to the **STS** or it SHOULD connect to a SOAP endpoint at the port `https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the STS domain name for a remote connection. Other port addresses MAY be used by implementations. [<3>](#)

3.3.4 Message Processing Events and Sequencing Rules

This port type utilizes the common message processing events and sequencing rules described in section [3.1.4](#).

3.3.5 Timer Events

This port type utilizes the common timer events design described in section [3.1.5](#).

3.3.6 Other Local Events

This port type utilizes the common other local events design described in section [3.1.6](#).

4 Protocol Examples

4.1 Issue Operation Examples

4.1.1 IssueRequest Example

This is an example of a message requesting issuance of a **SAML** token.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest</a:Action>
    <a:MessageID>urn:uuid:cc11441e-1d06-45b5-b0b5-ef73eee87659</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
    <a:To s:mustUnderstand="1">net.tcp://localhost/samlprotocol</a:To>
  </s:Header>
  <s:Body>
    <msis:IssueRequest
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:ActivityId>00000000-0000-0000-0000-000000000000</msis:ActivityId>
      <msis:Message>
        <msis:BaseUri>http://localhost</msis:BaseUri>

        <msis:SAMLRequest>PD94bWwgdmVyc2lvcj0iMS4wIiBlbmNvZGluZz0idXRmLTE2Ij8+PHNhbWxwOkFldGhuUmVxdWV
zdCBJRd0iX2QzYWNjZWl3LWVlZjctNDI5Ny1iMTgyLWE0NmYxYzQ3NWJjMSIgVmVyc2lvcj0iMi4wIiBjc3NlZUlu
bnQ9IjIwMDktMTItMTU0MDE2MDYyNDM0WiIgQ29uc2VudD0idXJuOm9hc2lzOm5hbWVzOnRjOlNBTUw6Mi4w
OmNvbNlbnQ6dW5zcGVjaWZpZWQIiHhtbG5zOnNhbmVwPSJlcm46b2FzaXM6bmFtZXM6dGM6U0FNTDoyLjA6
CHJvdG9jb2wiPjxJc3NlZXIgeGlsbnM9InVybjpvYXNpczpuYW1lc2p0YzptQU1MOjIuMDphc3NlcnRpb24i
Pmh0dHA6Ly9leHRlcm5hbHJwL3Njb3BlPC9uc3NlZXI+PC9zYW1scDpBdXRoblJlcXVlc3Q+</msis:SAMLRequest>

        <msis:PostBindingInformation></msis:PostBindingInformation>
      </msis:Message>
      <msis:OnBehalfOf>
        <wssc:SecurityContextToken wssu:Id=" 7b5d980c-9309-474e-ace8-23a99bbe261d-
6C82EA4288DB37210E653FCF8E064B57" xmlns:wssc="http://docs.oasis-open.org/ws-sx/ws-
secureconversation/200512" xmlns:wssu="http://docs.oasis-open.org/wss/2004/01/oasis-200401-
wss-wssecurity-utility-1.0.xsd">
          <wssc:Identifier>urn:uuid:24e876b6-1b0e-43e4-95da-7de16ec31f76</wssc:Identifier>
          <wssc:Instance>urn:uuid:a27fafd2-7e20-47c5-a004-3d83bed8e8f4</wssc:Instance>
          <mss:Cookie
xmlns:mss="http://schemas.microsoft.com/ws/2006/05/security">WFUABeuNCOWL9thXJ60luZ9/RNRXopMT
MYRhy/PRX3SAAAAABK91yIGlJLwXwgu5vEdh3wsm4zf7cBxsK5WaaM5TqQjGDlJ7qhgnjpNBwz9J7r/8fqJLdscGZvU7E
ifqfkkkoXX0IkDf+fUxXr0oBE/dY4BKGRk1SQ7VqOULAR4Xr39+X8Jp/eeMncIaJuZ01DSB4MwlulVpZKhC3grjfpFaOg
lwBwAAmoPlIv2HEhlhlpYbFBmaYmYzpqCOa/Ptr08YCN8YweH1FzEm929H5oEG87TMEjYnuAelBAMGo8BhgBtVS+o16jd
XCSeLF3J/vabemgbxIfJnqh4x5xuYldIRO9FJH78syGjOtGFAVi3KRnpIvnrPg3YKRW0sknIH21DDZjaFGPZW/wlB0Yen
bWFH+sRkfd+jOqhTVk+3++oeYcZWWSiAZhWDMZKA/kqv3Rh05Drr0v6JbzS3H+PJxzXL1NeEVd8Nhxh+0tINy+I3PWIHX
C7WFgYeS8TlTpaXBq+zrH5DDEQlV4haozU+41lT7pBcY9NdljLedSK/Eo5/Fyvm8g2HKL0jgKbr6jB3XYRFFDjAlTIWZ
jq7lZqqqAaaa9XzpcVKxZ4sHySUo1GK6uFYWpdZYUe5sTmW4bdYVJw6bNS0KEzhekLrocIe8c9Ldv61idHQJuvG0qT6xu
pCkD6KeUrV6ZQxRoKs5fyemG8sRw7R+p9tLpbPjqnpj4SbAjXwQJA0ksZ0KCDn+VBQIQ/YIc+fWd0Jv6/S+rZLDl1UXg
MYPmdGcfIMFZEMICjKNZ8IdcVGIXAuKC9AfFJpfgA+vQowgqXop6Abi//pKNrDa+ChNOQIkSFQoz5btiOpd63j5dKu/Y5
CqR+tHD7eYsrTflzdwhi0xYFAn0beoETCRSGsmCgo2iBvWWTxzelrKwPfn5wB0dnznH5ruPAOSQ9alto8k2dqbyavuPri
3MqehLnsBXR9Lh5j45gs19+InjbJv/Se1Xsbh5BkbTzP9pgkhG4ALivz1aRBcQdUDXe85Tb1hcJyDlAVs1PudCMHD1N8p
DDPAkgAzcIhTiBnElfljHL7uCeC+UKfEu6Hv8N134yw5vRPERgq8VaRBoUBXVSq9/p4Adv/HGTJbLU2hLl/rr9DqOru3h
Hlp0Rl8aLlDdHt/n004awqaIconXGILFqlwMRJXP3J8JL9CncPbp/eszX83o1GILPR+dnSRnAjHQbcGWkSM/5VmtHRvie
g1mXQVcJ90gltdmPictX8lUaxruGJezeIzkSwjtXHSg5HPEWGW91mtx9Sno4BD9XVeIMhKInBcyzKVlglr+AxopfNcGq
sKyA+EyGVOcscmDnqLmnR12gsvnYlJZBz2uBfgqCq6BLgIOqYNskdyEPTfcIn6aupftsh5zmnd/lvXY6b0TCXQ+iNuLm
bx1Ezh2znAL3UNqt4hDIQGEfwqR6TPKlp0dpd4T5yGtEcqOpfL2nwbCICsRLl1Snp4pBRuULw7cnlx4IzJcU+vplmGs
GpdtSUpJyxu+8XSAAh13wBxv8g+X3sZKNxKDAUncwHiq7QHHzPaRRat2S9i87+GJg6CFrIb32exctEY4c5eR/yXi8y2s
RTLqmf4X2s3+108sDMwcPunHh/ygRwK9NWq8BvuACpMkN5norSia+9//wyeeei9e3Ez3i/iMAWAyVoVYT1uom5jkwHEDR
FlZ0t5lRtejC1kPqFBAgDruJ+T402E3quHeGaRili7XRsY07EQocv07UGVOJ++YgtXb//SRdIFStO+MiOHv5AOIzDlab+
qKSRRhpSWmXK18x4Rja+5qBDE2+gPfj0lp42YC9ZSvrxHu/yHW/ZdNaaf106WAiaehYjIirfMiTx6yIXL0f6reF9FxpY
```


lvS2JQemJ5UWQJ1SVRiY1lZSXlpVFBKZ0UrNEkralIyT21leWVHemlzY0hZc2s3MG5wRWxGb1RKb2NXZXZYb3BTd28yRnZ
jNVF0V3dicHN4UnBXS3E4OCtjcXpuV0xoS0lZMG92Y2ZjNzVlaWFnM2xpK2NRajVESm5GS1pSenpJmZFoSupaRFJ3OXpM
TmR6eU8zN1J3RmVwRjhESTF6VDdwFIxSDJKV3ZNQW1nb29rSWH6ZDFXaEFDSHNNES4Q09nWnZENmh1d1BQYm9vSWNLt
XJYWmpwQkhXV1AxZG1pb3JVZ0hZa3czY0xkUzF4bTc5Rk9M2Z1JbWRMcmhSRFFZa0VxeWlRc1g5M2FBVHBtAnZvREgzMn
RSNg1Zc2tnYlMvOFJKAQRHmUxUXJ4dXlSsXQ1MmdrBDQvRwPxa1pZRHFXc0NQY1JxYWFxVTNybGJrUU1OT29sL2JvbU1
oenRFZi81dWl6UTFvY1AwV2J0UFVneXNPtNhtY29HQ3VISOxZcDBuRWkxdXhMdFN5R0RyQTJKeGhORnpHb0hraF16QlQy
WWHEM1ZmQ2x5YXR0N1R4OXRvT29qZUJ3bjYyWmxqMWhIdmNFUmFJRvV3VFhMznoxbmliVldiYURoV1ZlV2Nack14OXDIQ
TNZMGt2L3RJTE1qdlrL1dUTEM2d1RRcE16NXkzT1cyazNPdUZJTytmRWRCYT1XTNNb05HSENOwES1bEsxTDBmTGdYRG
NrYwxiTXNtNzRhaHE3L2xwZmJyU1FlcGdZY3ZCUnJsTmVBeFhGam15Z0F3Mv1RUHNBtM9FWkNJt2VEctJ0a1RqS01OeXh
kdEZTNitKbTBZ2JPb3FueGc3eJJKc1dwYVfocmtDcHg1LzFXbn11ZVRaMWJmV3cyaxZ4N3hnUjMxZXJlNUFTTGIwdUtH
U21LenVFSFA4NHfXr3N1OHA4Y0ExQmhoNzVXMFVBMG9Sc0o1Tzd5OVc3VXQ2dWY5R1hjcnhxcj1LLNzRlWnhJUDbVTEVsZ
lgzMXo5eW9pbW14WHNBZGtwR0p4bD1jc0ROeTdOcE1iOXBoZmpKTGFTSEFhYjhQcjI4U0h1NUU0L2ZrcVNxSk9kT2tJcX
JjUFZ4TV1Lb0pvaE9YtU84YmlINnhvdGZHODZREllcWo4ZS5TkvYNjv6UHg4cTBxVHY1b1JvQUFFV1YyUUVsc3daU2d
uTk1OSkRvVWxZMgd4RkNxaWduN05Kc0Q3WVYyWVZHDhntOVJmV2hgBhdON0czMMw1Vk05RlNrd0pEa3BLZVpsUU5tVHdm
WTZjVfJ5eXlZTlovQ3hHZ2IwYmdkL3hNZm42MG14Y1VldEtYVUhhkbjRvaG9nUXViUUPhM21Tt1FqS1JORG1EOEtOrmx3M
GFwd3pCYXyWt0tkU3NCQjAzUGN0YjVGWVZuNm5iNXVpTtNaW1YbFpOWsVaVdRbnVQT1Vya2hZaVpLNFR3SDBVU4rc3
V5YmNKNyt2dmlwenh1MkxLZjF3YisZajNwcXJTZnlBYjVzUW43OFpUSF1OUFPfYXU4RH1Ya0E0cEY3MH12SnVRL2tGS31
4WTVUUT44NnNBHRQekhaN2ZkU2dGVFRLEtE5IT09BeT1LzJIVWFEK2RBdmYramY5TVA3M31BajZnVmRzRXBETzNudjJ4
dXRiNHpCNktxSVdKbVNzK3ZkNEJqWDFLbE8xV0tXOUFRN0FaSkFzSTJQMzZhOW9PYVJwTkpITS9yaGJGK1VZcFMvY3pVW
Ut5NWVYWC9wb2pxellpR0s1VHg1OEXpDjMzM2R3ZWP5aGkZnm95NEg3K1NwWDJRYW1PNGVYTWRLR21SQ01vd0V6TFTLQV
lCOE9hY1NRSER4U2JVLytlUGlaNE9QZk5hMVV2RFJCBu43cVd3Tk1LSH1Rc0FDaWMwNE9YWU1CeW0zRE1weWx2SWUva2Z
RckFKbHBRNza0V3RU0UitlMk12R1Z4NEFTd1d1WHRs1Ndw1UWk1udzRmajhXUf1RdFk1SFZhuXvd2dW1DTmRL21a
WmVldUx4ajFBCFVMZUswK01YcTnHumlTYTJSYzBYQ2IxY3pLeVpXVE5FVmpORytLL2dxWEpSbH1WM1Zpd0JMK0Z5T09Lc
3hwT2JkamhHTHNYaTzqR2RWtYtMbTRKSOR2eTQrevRwa3F0K3JwTgd5M1VSSFJ3SERUWVNQC2NraG04TVAvbmSwT1ZLK2
JiN2pRZDd4bXbONFVrMXV2aStvcGZSS01FTVo3WjVubFgVskpWOC9DRXklDTVwMi96ZjVXZjR6eHpmSWExblp0WkxKVUz
oMnFoZUj1LzNEZjU1A3OV1UblpDdTz1WkU5RVVRMFRJUG1PdTBdVHo4cVM0VUpCL0tqQWV6Q3ZJQ0I2dn1OVndSOWh6
Ti9vazdVK2NLUHRsK1M3WfNRZDdpN0didFI5Slc2b1ZPQW1yVG05RGM5dTBibzFuNXpPTXpKYUUm1dnOEVG0VJoSTE0c
nAyQ3NpVdgyQnhFTnRyMXA3L1BtNndCQ1FBbWs5RUSaTdDcFNzdHRuOWpzaJjGWMnKd0o1REZIRGQzRWZveGRLMkw0Wk
5kdE9pSupaeEQ5bHVlQytSNHczT2V3b3pQWEVNdlnJenBirXQyUHNpdnR6VzFFVEYwZU5xcEt1R2FsSEhOUkRML0x6UGp
KYzd1S1BmUXZ5bHdp0TksZTEt1bm9rMkFTUUZyCvHCUgPWWNDNDandOemdOWFdadi9xUC9ue11UakpuUdJVFdBMITqNG1R
VWpCSFAxVFZzK3EvVDUwcCtsQ215Qys4Rml1T3Y2R0JNTG1Xd1d3QVFMNHFrV1VuRHU2bnUxSeX1MXZqdXZJcDjwTETYS
zhmNT1Lem1CdJfHU1VsYnBaY1BkVHJWSHcrcHNYRjNBW1RVU1d0TmPFMHpRY2pMUUVuc1hTRHN6S0dzVmh2d1BOQjhxL2
VvNnE0dTiZnNBGbE5tN0hzRjB3UHfWl0xtQVNZRlRNUDBkQitYMG1FTytLcWgxQzhqbKxkWWFVMjJrdCtGcStCbFhhSG1
MTSs0qlgvNnBhU2VSaU1PWkhjRkFsVEYxUU5iak1SaXdhbnJDQ1JpcHg4b1ZuVEDObWf2RzFVRnQzcnBucGFCRLpnMnNW
TWpha2kzZkpxUXkxbkVxejBLdHBUWFRaWURjK3I3U1M2TE02K1dXZitpTTR2Qkx1Ly9tejN0SD1haFpob0s1c0NedC9uZ
GJHSUk4emtYbngvai9aUXdtek9vMndwSFJmbXNYS25UMmNRNhp4dU51TzVUVVZTTmN1cko5dmhidTR3OTc2b2R4K0JJK2
1QZWtzL0hXtmxibXhYymx2czllcTVXWxVmUmhIVjFpd3NITWNKYThnZGdjY1lWN0NsaGRxUitPeVzSVWftbEpSemFEYtd
SWX1TbU1wbzZaT3BUEtDdhQWVNaWhnN1t4VFhMRVBDZHQJcZeDURUmJIRUpQbmJocEpXUFFkc21uWk1jYkZjREF5aFhu
VmdQVXJGYnIOVVRLbDFYFVdPcDNGUS94c1BHWGfVsmZRN0ZRYLZJNkdoUEFBamVPK1Y1Yw5Wa3NpekpTY0dsNXp5OE1GQ
1E1eEFsQ1pCSFJSLytNMWtvNzNNUnExSU8yZzhR0k1N3gyQWczZlhQsZJvRHVVTKdwVEc5c2JZRMv5VGxCTVBuRmtaTG
dLT1BhczZLZGEyUmNWZw02Q0tGeCt3bGU1TlQ5TQTd1dZGbZBLVEV3UF16QmdTa24zS2NgN0t5NmRUVGxIaUtjU0QxaCt
Pd2FY1bWzZaT3BUEtDdhQWVNaWhnN1t4VFhMRVBDZHQJcZeDURUmJIRUpQbmJocEpXUFFkc21uWk1jYkZjREF5aFhu
OULzekhhL1YrOVdFMnRVZklqQ1JNRy8venV2WT08L3h1bmM6Q2lwaGvYVmFsdWU+PC94Zw5jOkNpcGhlckRhDGE+PC94Z
W5jOkVuY3J5cHRlZERhdGE+PC9FbmNyeXB0ZWRC3N1cnRpb24+PC9zYw1scDpSZXNwb25zZT4=</msis:SAMLRespons
e>

```
<msis:PostBindingInformation></msis:PostBindingInformation>
</msis:Message>
<msis:SessionState></msis:SessionState>
<msis:AuthenticatingProvider>http://localhost/</msis:AuthenticatingProvider>
</msis:IssueResponse>
</s:Body>
</s:Envelope>
```

4.1.3 IssueResponse Example Using Artifact Binding

This is an example of a reply to a request to issue a **SAML** token, which contains the resulting SAML response message. In this example, the **SAML Artifact Binding** was employed.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:RelatesTo urn:uuid:0ac7deb2-4d52-4a77-8071-d4bb099e6db9</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
```


4.2.2 CreateErrorMessageResponse Example

This is an example of a reply to a message requesting creation of a **SAML** error message, which contains the created SAML error message.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:678452fe-e24d-439e-8543-e2e72f936930</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:CreateErrorMessageResponse
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:Message>
        <msis:BaseUri>https://externalrp</msis:BaseUri>

<msis:SAMLResponse>7b0HYBxJliUmL23Ke39K9UrX4HShCIBgEyTYkEAQ7MGIZEaS7B1pRyMpqqyqBymVWZV1mFkDM7Z
28995777333nvvvfe60510J/ff/z9cZmQBbPb0StrJniGAqsgfP358Hz8iHjfZolw9epU3q2rZ5OnZ088++v2zLL+/vzO
5tz3Zy/e29/PJ/el5z7q7ne8/yPbuP9g9mB3MPkp/Mq+bolp+9tHeeOej9KxplvnZsmmzZUzf7ew83N7d2949eLOz++je
7qPd3fHDvb2f+ih9mjdtscafnPetqvm0d27+bs2r5dZwa/ufpSeABEAwdfLR1XWFM2jZbbIm0ft9Nhr4y+eP6L+Hk2l0
aPlslnl0+K8yAmjs6UZx5uKhrG38yD/NHuQbe/czw6296f3Hm5Pdnf3tw/2pg8e5g/vT6fnBx+17xblsnnEZnJc5aqu2m
palR8dPebBlvLq5peypslrDPajIwyWxlpW06ycV0179/FdgXOkk/C6zdp1E/51Us3y9Cezcplv7qfh1jqPs7z+6IPAvKi
01+18STPR0tR81N49eny3C67zkf3TTMHR/wM=</msis:SAMLResponse>
        <msis:RedirectBindingInformation>

<msis:Signature>RlFtupsaiITbNa5wL4+mOnuFpRBYs5kq/ni5ycqNprqpol0c5+RUOA5/8RkmRY787oB817FfFJOYw
3FkIlhWayPqc1b1HFp7AcuJFPmWVT2bGXbdRV6sCFV0g5XOlPsYG+a/9EZdiYUaMCRUvOds0s5SdtmL95FCQpLxkG5PEk
w=</msis:Signature>
        <msis:SigAlg>http://www.w3.org/2001/04/xmldsig-more#rsa-sha256</msis:SigAlg>
      </msis:RedirectBindingInformation>
    </msis:Message>
  </msis:CreateErrorMessageResponse>
</s:Body>
</s:Envelope>
```

4.3 SignMessage Operation Examples

4.3.1 SignMessageRequest Example

This is an example of a message that requests that a **SAML Message** signature be applied to a SAML Message.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest</a:Action>
    <a:MessageID>urn:uuid:5654c3f9-691f-4f9e-aa51-d5d37060dc88</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
    <a:To s:mustUnderstand="1">net.tcp://localhost/samlprotocol</a:To>
  </s:Header>
  <s:Body>
    <msis:SignMessageRequest
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:ActivityId>00000000-0000-0000-0000-000000000000</msis:ActivityId>
```



```

<msis:Message>
  <msis:BaseUri>http://contoso.com/</msis:BaseUri>

  <msis:SAMLRequest>PHNhbWxwOkFlZGhuUmVxdWVzdCBJRd0iXzA4MTZjZjJiLTg2YzUtNDU2Ny04MGVlLTfKzjVmYjV
  jZmYzYiIgVmVyc2l2b2J0iMi4wIiBjc3NlZUlu3Rhb3N0OjQzNDMvbnVuaXQvRmVhZG1vblBhc3NpdUvIiBD25zZW50PSJlcm46b2Fza
  b249Imh0dHBzOi8vbG9jYXRob3N0OjQzNDMvbnVuaXQvRmVhZG1vblBhc3NpdUvIiBD25zZW50PSJlcm46b2Fza
  XM6bmFtZXM6dGM6U0FNTDoyLjA6Y29uc2VudDp1bnNwZWNPZml1ZCIGeGlsbnM6c2FtbHA9InVybjpvczpuYWllcz
  p0YzptQUlMOjIuMDpwcm90b2NvbCIgZ4=</msis:SAMLRequest>
  <msis:PostBindingInformation></msis:PostBindingInformation>
</msis:Message>
<msis:Principal>
  <msis:Type>Scope</msis:Type>
  <msis:Identifier>http://externalrp/rpl</msis:Identifier>
</msis:Principal>
</msis:SignMessageRequest>
</s:Body>
</s:Envelope>

```

4.3.2 SignMessageResponse Example

This is an example of a reply to a request to create a signed **SAML Message**, which contains the resulting SAML Message.

```

<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
  xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
      s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/Proc
      essRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:5654c3f9-691f-4f9e-aa51-d5d37060dc88</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:SignMessageResponse
      xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:Message>
        <msis:BaseUri>http://contoso.com/</msis:BaseUri>

        <msis:SAMLRequest>PHNhbWxwOkFlZGhuUmVxdWVzdCBJRd0iXzA4MTZjZjJiLTg2YzUtNDU2Ny04MGVlLTfKzjVmYjV
        jZmYzYiIgVmVyc2l2b2J0iMi4wIiBjc3NlZUlu3Rhb3N0OjQzNDMvbnVuaXQvRmVhZG1vblBhc3NpdUvIiBD25zZW50PSJlcm46b2Fza
        XM6bmFtZXM6dGM6U0FNTDoyLjA6Y29uc2VudDp1bnNwZWNPZml1ZCIGeGlsbnM6c2FtbHA9InVybjpvczpuYWllcz
        p0YzptQUlMOjIuMDpwcm90b2NvbCIgZ4=</msis:SAMLRequest>
        <msis:PostBindingInformation></msis:PostBindingInformation>
      </msis:Message>
    </s:Body>
  </s:Envelope>

```



```

lcnRpZmljYXRlPjwvZHM6WDUwOURhdGE+PC9LZXlJbmZvPjwvZHM6U2lnbmF0dXJlPjwvc2FtbHA6QXV0aG5SZXF1ZXN0
Pg==</msis:SAMLRequest>
  <msis:PostBindingInformation></msis:PostBindingInformation>
</msis:Message>
</msis:VerifyMessageRequest>
</s:Body>
</s:Envelope>

```

4.4.2 VerifyMessageResponse Example

This is an example of a reply to a request to verify that a **SAML Message** is from a known party and signed according to the configuration for that party, containing the verification response.

```

<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:05fbb0db-e105-448b-b127-1bf689738d75</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:VerifyMessageResponse
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:IsVerified>true</msis:IsVerified>
    </msis:VerifyMessageResponse>
  </s:Body>
</s:Envelope>

```

4.4.3 VerifyMessageResponse Example Using Redirect Binding

This is an example of a reply to a request to verify that a **SAML Message** is from a known party and signed according to the configuration for that party, containing the verification response. In this example, the **SAML redirect binding** was employed.

```

<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequest</a:Action>
    <a:MessageID>urn:uuid:9525a87c-7b8b-474b-9840-eb3f202ab467</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
    <a:To s:mustUnderstand="1">net.tcp://localhost/samlprotocol</a:To>
  </s:Header>
  <s:Body>
    <msis:VerifyMessageRequest
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:ActivityId>00000000-0000-0000-0000-000000000000</msis:ActivityId>
      <msis:Message>
        <msis:BaseUri>http://contoso.com/</msis:BaseUri>

<msis:SAMLRequest>7b0HYBxJliUmL23Ke39K9UrX4HShCIBgEyTYkEAQ7MGIzeaS7B1pRyMpqqqBymVWZV1mFkDM7Z2
8995777333nvvvfe60510J/ff/z9cZmQBbPbOstrJniGAqsgfP358Hz8iHjZolw90l638+Wr/Bet86ZNz55+9tHvv30w
++n0fG+yffDp9P72/v1PH2wf70T59u7s/P755P70/Pze5KP0J/O6KarlZx/tjXc+Ss+aZp2fLZs2W7b00c70w+3dve3dg
zc7u4/u7T7avTd+sLv7Ux+lT6mXYpm1/Oa8bVfNo7t3y2qalfOqaR/t39u/d3e5Xhbt3Wf5LK+54cusaYrL/O5H6Um1bH
J0sK6Xj6qsKZpHy2yRN4/a6aPXx188f0S4PJpKo0frZbPKp8V5kc8+St8tymXziEe8+e1VXbXVtCo/OnrMY6r11c0vEYJ
5DVQ/OsKY/CHdfXxX4Bw9vtSn/NH/Aw==</msis:SAMLRequest>

```



```

yMlJpekFnTUJBQUdQl1RCSElFVUdBMVVKQVFRK01EeUffS2tpemh00ExGNkxtbWt3ZXd5V3dUcWhGakFVTvJJd0VBWURW
UVFERXdsC2IyTmhiR2h2YzNTQ0VFU1NnWHPjY1Vtb1FnUVFWb1pmd05Bd0RRWUpLb1pJaHZjTkFRRUVCUUFZE11lFQVBUN
XhBNHc4UHR0SGwvZmtOLgZV93R1L2NWUfQYkpNVmdDbDV5TGZRMUxONmdrek4veFVPcHhVZUY1eXF0Tz1PdXRWRXVfD2
Zzald4cHlxZVCT09ZeFVGdDBRS1lvVnNyeDA4MldETktpbkhiVXlYdkVKLz1Eb2IrdCtpU1BqdTdlbVJKYVRLbm52bWh
HUH10WnU1c2s5clQ3TzEwSEpTQkJGSEkyRmZkQT08L2RzOlglMDlDZXJ0aWZpY2F0ZT48L2RzOlglMDlEYXRhPjwvS2V5
SW5mbz48L2RzOlNpZ25hdHVyZT48TmFtZU1EIHhtbG5zPSJlcm46b2FzaXM6bmFtZXM6dGM6U0FNTDoyLjA6YXNzZXJ0a
W9uIj5iYXI8L05hbWVJRd48c2FtbHA6U2Vzc21vbkluzGV4PjAwMDwvc2FtbHA6U2Vzc21vbkluzGV4Pjwvc2FtbHA6TG
9nb3V0UmVxdWVzdD4=</msis:SAMLRequest>
    <msis:PostBindingInformation></msis:PostBindingInformation>
  </msis:Message>
  <msis:SessionState>http%3a%2f%2flocalhost%2f&True&aaa&&&&111</msis:SessionState>

<msis:LogoutState>http%3a%2f%2fexternalrp%2fscope?_ID??http%3a%2f%2fexternalrp%2fscope&False&
foo&&&&?ID?</msis:LogoutState>
</msis:LogoutRequest>
</s:Body>
</s:Envelope>

```

4.5.2 LogoutResponse Example

This is an example of a reply to a request that a **SAML** logout be performed, which contains the updated SessionState and LogoutState values.

```

<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
  xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
      s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/Proc
      essRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:17817720-c31e-48e8-8904-067aac199c8d</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:LogoutResponse
      xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:LogoutStatus>InProgress</msis:LogoutStatus>
      <msis:Message>
        <msis:BaseUri>https://localhost:4343/SLO/RedirectResponse</msis:BaseUri>

        <msis:SAMLResponse>7b0HYBxJliUmL23Ke39K9UrX4HShCIBgEyTYkEAQ7MGIzeaS7B1pRyMpqqyqBymVWZV1mFkDM7Z
        28995777333nvvvfe6051OJ/ff/z9cZmQBbPbOStrJniGAqsgfP358Hz8iHjfZolw9e15dVov2Vd6sqmWTP2dPP/vo958
        9zD69n+/vbM8e7j3Y3t+/v7N9cJDvbO/s3N+f3D+/n+2cTz5KfzKvm6JafvBR3njno/Ssadb52bJps2VLH+3sPNze3dve
        PXizs/vo3u6jvb3xg4cPfuqj9GnetMUya/nNeduumkd375bVNCvnVdM+2r+3f+/u6+df3n2Vz4o6n1rMPkpP8APA1/XyU
        ZU1RfNomS3y51E7ffT6+IvnjwiPR1Np9Gi9bFb5tDgv8hnhjtRQ3lQ0vIMH53t7+d6n2+e7D3a29w9m+9sHuzu72/mDWX
        bw6e79Sf6Qunu3KJfNIybS5i5XddVW06r86Ogx6GWVze/1DVNXoMIHx2BCD4N7j6+K3COdIpetlm7bsK/TqpZnv5kVq7
        zzf003PrRq/wXrYnweflRevfo8d0Qrv4ZcsLR/wM=</msis:SAMLResponse>
      <msis:RedirectBindingInformation>

      <msis:Signature>AIN+zc9QDY7YZ65zRXz0ob4RMuElAGEPuok37NCdWvubEJ4E3awvi8Ieu+v+LsDhBd+zXZmjB7NDU
      XUcoTzqloFN0Whlbq34OrMitR4FbGDQMpwBylVlmy2MXN7nZvAD+2en+Pd+bkk4P0KMH7PPCQsboj63CyzRfGnV+R81Mf
      Y=</msis:Signature>
      <msis:SigAlg>http://www.w3.org/2001/04/xmldsig-more#rsa-sha256</msis:SigAlg>
    </msis:RedirectBindingInformation>
  </msis:Message>
  <msis:SessionState>http%3a%2f%2flocalhost%2f&True&aaa&&&&111</msis:SessionState>

  <msis:LogoutState>http%3a%2f%2fexternalrp%2fscope?_ID??http%3a%2f%2fexternalrp%2fscope&False&
  foo&&&&?ID?</msis:LogoutState>
</msis:LogoutResponse>
</s:Body>
</s:Envelope>

```

4.5.3 LogoutRequest Example - Locally Initiated

This is an example of a message requesting that a **SAML** logout be performed. In this example, the request is being sent to the endpoint on the local host.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:MessageID>urn:uuid:1fec3465-1008-490d-aeb2-da9b4df4a3d2</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
    <a:To s:mustUnderstand="1">net.tcp://localhost/samlprotocol</a:To>
  </s:Header>
  <s:Body>
    <msis:LogoutRequest
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:ActivityId>00000000-0000-0000-0000-000000000000</msis:ActivityId>
      <msis:SessionState></msis:SessionState>
      <msis:LogoutState></msis:LogoutState>
    </msis:LogoutRequest>
  </s:Body>
</s:Envelope>
```

4.5.4 LogoutResponse Example:Final Response to Locally Initiated Request

This is an example of a reply to a request that a **SAML** logout be performed, which contains the updated SessionState and LogoutState values. In this example, the final response to a locally initiated logout request is shown.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:1fec3465-1008-490d-aeb2-da9b4df4a3d2</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:LogoutResponse
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:LogoutStatus>LogoutSuccess</msis:LogoutStatus>
      <msis:SessionState></msis:SessionState>
      <msis:LogoutState></msis:LogoutState>
    </msis:LogoutResponse>
  </s:Body>
</s:Envelope>
```

4.5.5 LogoutRequest Example with SAMLResponse and RelayState

This is an example of a message requesting that a **SAML** logout be performed. In this example, the request contains both a SAMLResponse and RelayState.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
```


4.5.6 LogoutResponse Example with SAMLRequest and RelayState

This is an example of a reply to a request that a **SAML** logout be performed, which contains the updated SessionState and LogoutState values. In this example, the response is to a request where the request contains both a SAMLResponse and RelayState.

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action
s:mustUnderstand="1">http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/ProcessRequestResponse</a:Action>
    <a:RelatesTo>urn:uuid:9aaa9e19-93b7-46e5-afb8-24804341d435</a:RelatesTo>
    <a:To s:mustUnderstand="1">http://www.w3.org/2005/08/addressing/anonymous</a:To>
  </s:Header>
  <s:Body>
    <msis:LogoutResponse
xmlns:msis="http://schemas.microsoft.com/ws/2009/12/identityserver/samlprotocol/">
      <msis:LogoutStatus>InProgress</msis:LogoutStatus>
      <msis:Message>
        <msis:BaseUri>https://externalrp:4343/SLO/RedirectRequest</msis:BaseUri>

<msis:SAMLRequest>7b0HYBxJliUmL23Ke39K9UrX4HShCIBgEyTYkEAQ7MGIzeaS7BlpRyMpqqyqBymVWZVlmFkDM7Z2
8995777333nvvvfe60510J/ff/z9cZmQBbPbOStrJniGAqsgfP358Hz8iHjfZolw9e15dVOv2Vf6LlnnTpm DPP/v0988f
5vd39/J8e+fBwWx7//7+dPvhvXyyPdmDZg/vH0WeTrL7H6U/mddNUS0/+2hvvPNRetY06/xs2bTZsqWPdnYebu/ube8ev
NnZfXRv99HevfG9vYOf+ih9Sr0Uy6z1N+dtu2oe3b2bv2vzepmV9erR/r39e3dfP//y7qt8VtT51GD2UXpSLZscwNf18l
GVNUXzaJkt8uZRO330+viL548Ij0dTafRovWxW+bQ4L/LZR+mLqvly+WV9fE699HD71OH2b1Eum0dMls29rOqgraZV+dH
RYx53La9ufilemrzGuD86wrhp2Gulzcp51bR3H98VOEePX9B7Z0/fE955VT2+K28e6bS+zhvMztlylr872tnZeXw38rn5
MOCBo/8H</msis:SAMLRequest>
      <msis:RedirectBindingInformation>
        <msis:RelayState>RelayState</msis:RelayState>

<msis:Signature>TgTFsKkfCEEtm6iul8kZzRzxOOqCxAgelkobQaaS6vV8iXeqmIAdYBvZeTykQaif3KYp5herI6evS
MXA1P7KwX/GG/8o5e6QbNiBZTn48Cti+YJF7yqCZ5HPX/gRg9e9CL8LvMvy8hBa8rDnDOH3eRZFwQNSzJzdVSqs+TNAX+
4=</msis:Signature>
      <msis:SigAlg>http://www.w3.org/2001/04/xmldsig-more#rsa-sha256</msis:SigAlg>
    </msis:RedirectBindingInformation>
  </msis:Message>
  <msis:SessionState></msis:SessionState>

<msis:LogoutState>http%3a%2f%2fexternalrp%2fscope?ID??http%3a%2f%2fexternalrp%2fscope&False&f
oo&&&&000? e9e512ee-078d-454c-93eb-
b1ca958b9ba5?urn%3aoasis%3anames%3atc%3aSAML%3a2.0%3astatus%3aSuccess</msis:LogoutState>
    </msis:LogoutResponse>
  </s:Body>
</s:Envelope>
```

5 Security

5.1 Security Considerations for Implementers

Implementers have to ensure that SSL is used to authenticate between clients and servers on different machines, and that the server is the intended server referred to by the server endpoint. Implementers also have to ensure that the remote client role authenticates to the server role such that the server can trust the client to perform SSL client **certificate** authentication where appropriate. Otherwise there are no specific security considerations beyond those specified in normative references.

5.2 Index of Security Parameters

None.

6 Appendix A: Full WSDL

For ease of implementation, the following example provides the full **Web Services Description Language (WSDL)** ([\[WSDL\]](#)).

```
<?xml version="1.0" encoding="utf-8"?>
<wsdl:definitions xmlns:wsa10="http://www.w3.org/2005/08/addressing"
xmlns:wsx="http://schemas.xmlsoap.org/ws/2004/09/mex"
xmlns:soap12="http://schemas.xmlsoap.org/wsdl/soap12/" xmlns:wsu="http://docs.oasis-
open.org/wss/2004/01/oasis-200401-wss-wssecurity-utility-1.0.xsd"
xmlns:wsp="http://schemas.xmlsoap.org/ws/2004/09/policy"
xmlns:wsap="http://schemas.xmlsoap.org/ws/2004/08/addressing/policy"
xmlns:msc="http://schemas.microsoft.com/ws/2005/12/wsdl/contract"
xmlns:wsa="http://schemas.xmlsoap.org/ws/2004/08/addressing"
xmlns:wsam="http://www.w3.org/2007/05/addressing/metadata"
xmlns:wsaw="http://www.w3.org/2006/05/addressing/wsdl" xmlns:tns="http://tempuri.org/"
xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
targetNamespace="http://tempuri.org/" xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/">
  <wsdl:types />
  <wsdl:portType name="ISamlProtocolContract" />
  <wsdl:portType name="IAnyActionContract" />
  <wsdl:binding name="DefaultBinding_ISamlProtocolContract" type="tns:ISamlProtocolContract">
    <soap:binding transport="http://schemas.xmlsoap.org/soap/http" />
  </wsdl:binding>
  <wsdl:binding name="DefaultBinding_IAnyActionContract" type="tns:IAnyActionContract">
    <soap:binding transport="http://schemas.xmlsoap.org/soap/http" />
  </wsdl:binding>
</wsdl:definitions>
```

7 Appendix B: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows Server 2003 R2 operating system
- Windows Server 2008 operating system
- Windows Server 2008 R2 operating system
- Active Directory Federation Services (AD FS) 2.0
- Windows Server 2012 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 3.1.3](#): AD FS 2.0 does use the **SOAP** endpoint address `net.tcp://localhost/samlprotocol` to establish local connections and the SOAP endpoint address

`https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the **STS** server domain name, to establish remote connections.

[<2> Section 3.2.3](#): AD FS 2.0 does use the SOAP endpoint address `net.tcp://localhost/samlprotocol` to establish local connections and the SOAP endpoint address

`https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the STS server domain name, to establish remote connections.

[<3> Section 3.3.3](#): AD FS 2.0 does use the SOAP endpoint address `net.tcp://localhost/samlprotocol` to establish local connections and the SOAP endpoint address

`https://contoso.com/adfs/services/trust/samlprotocol/proxycertificatetransport`, where `contoso.com` represents the STS server domain name, to establish remote connections.

8 Change Tracking

No table of changes is available. The document is either new or has had no changes since its last release.

9 Index

A

Abstract data model
client ([section 3.1.1](#) 23, [section 3.3.1](#) 31)
server ([section 3.1.1](#) 23, [section 3.2.1](#) 30)

[Applicability](#) 10
[Attribute groups](#) 22
[Attributes](#) 22

C

[Capability negotiation](#) 10
[Change tracking](#) 49
Client
abstract data model ([section 3.1.1](#) 23, [section 3.3.1](#) 31)
[CreateErrorMessage operation](#) 25
initialization ([section 3.1.3](#) 23, [section 3.3.3](#) 31)
[Issue operation](#) 25
local events ([section 3.1.6](#) 30, [section 3.3.6](#) 32)
[Logout operation](#) 25
message processing ([section 3.1.4](#) 23, [section 3.3.4](#) 31)
[multiple operations](#) 26
overview ([section 3.1](#) 23, [section 3.3](#) 31)
sequencing rules ([section 3.1.4](#) 23, [section 3.3.4](#) 31)
[SignMessage operation](#) 24
timer events ([section 3.1.5](#) 30, [section 3.3.5](#) 31)
timers ([section 3.1.2](#) 23, [section 3.3.2](#) 31)
[VerifyMessage operation](#) 24
[Complex types](#) 19
[overview](#) 19
[PostBindingType](#) 20
[PrincipalType](#) 19
[RedirectBindingType](#) 21
[RequestType](#) 19
[ResponseType](#) 19
[SamlMessageType](#) 20
[CreateErrorMessage operation](#) 25
[CreateErrorMessageRequest example](#) 36
[CreateErrorMessageRequest message](#) 18
[CreateErrorMessageResponse example](#) 37
[CreateErrorMessageResponse message](#) 18

D

Data model - abstract
client ([section 3.1.1](#) 23, [section 3.3.1](#) 31)
server ([section 3.1.1](#) 23, [section 3.2.1](#) 30)

E

Events
local
client ([section 3.1.6](#) 30, [section 3.3.6](#) 32)
server ([section 3.1.6](#) 30, [section 3.2.6](#) 31)
[local - client](#) 32
[local - server](#) 31
timer
client ([section 3.1.5](#) 30, [section 3.3.5](#) 31)
server ([section 3.1.5](#) 30, [section 3.2.5](#) 31)

[timer - client](#) 31
[timer - server](#) 31

Examples

[CreateErrorMessageRequest](#) 36
[CreateErrorMessageResponse](#) 37
[IssueRequest](#) 33
[IssueResponse](#) 34
[IssueResponse example using artifact binding](#) 35
[LogoutRequest](#) 41
[LogoutRequest example - locally initiated](#) 43
[LogoutRequest example with SAMLResponse and RelayState](#) 43
[LogoutResponse](#) 42
[LogoutResponse example - final response to locally initiated request](#) 43
[LogoutResponse example with SAMLRequest and RelayState](#) 45
[SignMessageRequest](#) 37
[SignMessageResponse](#) 38
[VerifyMessageRequest](#) 39
[VerifyMessageResponse](#) 40
[VerifyMessageResponse example using redirect binding](#) 40

F

[Fields - vendor-extensible](#) 11
[Full WSDL](#) 47

G

[Glossary](#) 7
[Groups](#) 22

I

[Implementer - security considerations](#) 46
[Index of security parameters](#) 46
[Informative references](#) 9
Initialization
client ([section 3.1.3](#) 23, [section 3.3.3](#) 31)
server ([section 3.1.3](#) 23, [section 3.2.3](#) 31)
[Introduction](#) 7
[Issue operation](#) 25
[IssueRequest example](#) 33
[IssueRequest message](#) 15
[IssueResponse example](#) 34
[IssueResponse example using artifact binding](#) 35
[IssueResponse message](#) 16

L

Local events
client ([section 3.1.6](#) 30, [section 3.3.6](#) 32)
server ([section 3.1.6](#) 30, [section 3.2.6](#) 31)
[Logout operation](#) 25
[LogoutRequest example](#) 41
[LogoutRequest example - locally initiated](#) 43
[LogoutRequest example with SAMLResponse and RelayState](#) 43
[LogoutRequest message](#) 16
[LogoutResponse example](#) 42

[LogoutResponse example - final response to locally initiated request](#) 43
[LogoutResponse example with SAMLRequest and RelayState](#) 45
[LogoutResponse message](#) 17
[LogoutStatusType simple type](#) 21

M

Message processing
 client ([section 3.1.4](#) 23, [section 3.3.4](#) 31)
 server ([section 3.1.4](#) 23, [section 3.2.4](#) 31)
Messages
 [attribute groups](#) 22
 [attributes](#) 22
 [complex types](#) 19
 [CreateErrorMessageRequest](#) 18
 [CreateErrorMessageRequest message](#) 18
 [CreateErrorMessageResponse](#) 18
 [CreateErrorMessageResponse message](#) 18
 [elements](#) 19
 enumerated ([section 2.2.2](#) 12, [section 2.2.2.1](#) 13)
 [groups](#) 22
 [IssueRequest](#) 15
 [IssueRequest message](#) 15
 [IssueResponse](#) 16
 [IssueResponse message](#) 16
 [LogoutRequest](#) 16
 [LogoutRequest message](#) 16
 [LogoutResponse](#) 17
 [LogoutResponse message](#) 17
 [LogoutStatusType simple type](#) 21
 [namespaces](#) 12
 [PostBindingType complex type](#) 20
 [PrincipalType complex type](#) 19
 [PrincipalTypes simple type](#) 22
 [RedirectBindingType complex type](#) 21
 [RequestType complex type](#) 19
 [ResponseType complex type](#) 19
 [SamlMessageType complex type](#) 20
 [SignMessageRequest](#) 13
 [SignMessageRequest message](#) 13
 [SignMessageResponse](#) 14
 [SignMessageResponse message](#) 14
 [simple types](#) 21
 [syntax](#) 12
 [transport](#) 12
 [VerifyMessageRequest](#) 14
 [VerifyMessageRequest message](#) 14
 [VerifyMessageResponse](#) 15
 [VerifyMessageResponse message](#) 15
[Multiple operations](#) 26

N

[Namespaces](#) 12
[Normative references](#) 8

O

Operations
 [CreateErrorMessage](#) 25
 [Issue](#) 25
 [Logout](#) 25
 [multiple operations](#) 26

[SignMessage](#) 24
[VerifyMessage](#) 24
[Overview \(synopsis\)](#) 9

P

[Parameters - security index](#) 46
[PostBindingType complex type](#) 20
[Preconditions](#) 10
[Prerequisites](#) 10
[PrincipalType complex type](#) 19
[PrincipalTypes simple type](#) 22
[Product behavior](#) 48

R

[RedirectBindingType complex type](#) 21
[References](#) 8
 [informative](#) 9
 [normative](#) 8
[Relationship to other protocols](#) 9
[RequestType complex type](#) 19
[ResponseType complex type](#) 19

S

[SamlMessageType complex type](#) 20
Security
 [implementer considerations](#) 46
 [parameter index](#) 46
Sequencing rules
 client ([section 3.1.4](#) 23, [section 3.3.4](#) 31)
 server ([section 3.1.4](#) 23, [section 3.2.4](#) 31)
Server
 abstract data model ([section 3.1.1](#) 23, [section 3.2.1](#) 30)
 [CreateErrorMessage operation](#) 25
 initialization ([section 3.1.3](#) 23, [section 3.2.3](#) 31)
 [Issue operation](#) 25
 local events ([section 3.1.6](#) 30, [section 3.2.6](#) 31)
 [Logout operation](#) 25
 message processing ([section 3.1.4](#) 23, [section 3.2.4](#) 31)
 [multiple operations](#) 26
 [overview](#) 23
 sequencing rules ([section 3.1.4](#) 23, [section 3.2.4](#) 31)
 [SignMessage operation](#) 24
 timer events ([section 3.1.5](#) 30, [section 3.2.5](#) 31)
 timers ([section 3.1.2](#) 23, [section 3.2.2](#) 30)
 [VerifyMessage operation](#) 24
[SignMessage operation](#) 24
[SignMessageRequest example](#) 37
[SignMessageRequest message](#) 13
[SignMessageResponse example](#) 38
[SignMessageResponse message](#) 14
[Simple types](#) 21
 [LogoutStatusType](#) 21
 [overview](#) 21
 [PrincipalTypes](#) 22
[Standards assignments](#) 11
Syntax
 [messages - overview](#) 12
[Syntax - messages - overview](#) 12

T

Timer events

- client ([section 3.1.5](#) 30, [section 3.3.5](#) 31)
- server ([section 3.1.5](#) 30, [section 3.2.5](#) 31)

Timers

- client ([section 3.1.2](#) 23, [section 3.3.2](#) 31)
- server ([section 3.1.2](#) 23, [section 3.2.2](#) 30)

[Tracking changes](#) 49

[Transport](#) 12

Types

- [complex](#) 19
- [simple](#) 21

V

[Vendor-extensible fields](#) 11

[VerifyMessage operation](#) 24

[VerifyMessageRequest example](#) 39

[VerifyMessageRequest message](#) 14

[VerifyMessageResponse example](#) 40

[VerifyMessageResponse example using redirect
binding](#) 40

[VerifyMessageResponse message](#) 15

[Versioning](#) 10

W

[WSDL](#) 47